

Solutions — Problem Set 7

MACROECONOMIA 1 — MPE Insper — Prof. Pedro G. Duarte

2026, 3º trimestre · Due: 25/09

The New-Keynesian AS-AD model — Benigno (2015), §§1–12

How to read this document

This file answers **all four questions** (10 points). No textbook exercise sits behind this list: every item is built on Benigno (2015), “New-Keynesian economics: an AS-AD view”, *Research in Economics* 69, printed pages 503–524, which is the reading for lectures 8 and 9.

The thread. The four questions are one argument told four times. Question 1 asks *why* the New-Keynesian model has the ingredients it has. Question 2 uses them: one shock that moves AS, one policy that moves AD. Question 3 isolates the single ingredient — the mark-up wedge — that makes the natural and the efficient level of output different objects. Question 4 shows what a central bank does once they are different: it minimises a welfare loss, and the AS curve is the menu it must choose from.

The model, once. Every answer below uses only these lines (Benigno’s notation: lower-case letters are log-deviations from an efficient steady state):

$$\begin{aligned} \text{AS (17), (20): } p - p^e &= \kappa (y - y_n), & \kappa &\equiv \frac{(1 - \alpha)(\sigma^{-1} + \eta)}{\alpha}, \\ \text{AD (21): } y &= \bar{y}_n + (g - \bar{g}) - \sigma [i - (\bar{p} - p) - (\bar{\tau}_c - \tau_c) - \rho], \\ \text{natural (15): } y_n &= \frac{(1 + \eta)a + \sigma^{-1}g - \mu}{\sigma^{-1} + \eta}, & \text{efficient (19): } y_e &= \frac{(1 + \eta)a + \sigma^{-1}g}{\sigma^{-1} + \eta}. \end{aligned}$$

Numbers use Benigno’s own calibration: $\alpha = 0.66$, $\sigma = 0.5$, $\eta = 0.2$ (p. 515), so $\kappa = 1.133$; and $\theta = 8$ (fn. 24, p. 522). They are computed in `Resolucao/lista7_codigo/17_figuras.py`, which prints every figure quoted here.

Companions. Three interactive pages were built for this list. Each has its own controls and its read-outs are the closed forms below, evaluated live: [Two Shocks](#) (Q2), [The Wedge](#) (Q3), [The Loss Bowl](#) (Q4). Local copies are in `Map/benigno/`. The full derivation of every equation is in the note set `Map/benigno/00-index.md`.

Highlighted answers are in **blue**. **Red** boxes are traps; **green** boxes are method. *Language:* English, per the project’s global output rule; the problem set is itself in English.

Item	Points	Basis in Benigno (2015)
1 — NK vs New Classical vs Keynesian	3	§1–§4, §11; pp. 503–508, 521–522
2(a) — temporary mark-up rise	1.5	§7, Fig. 9, pp. 513–514
2(b) — nominal-rate cut	1.5	§5, Figs. 3–5, pp. 509–511
3 — y_n vs y_e	2	§4.1–§4.4, eqs. (14), (15), (19), pp. 507–509
4 — the loss function and the AS constraint	2	§11, eqs. (33)–(35), Fig. 14, pp. 521–523

Question 1 Three generations of short-run macroeconomics

The one-sentence answer, before the table

The New-Keynesian (NK) model takes its **method** from the New Classics (optimising agents, rational expectations, general equilibrium) and its **frictions** from the Keynesians (nominal prices that do not clear markets in the short run). Two ingredients turn that combination into a working model: **monopolistic competition**, so that someone *sets* the prices that are sticky, and an **interest-rate**

instrument, so that demand is driven by the real rate through the Euler equation. Benigno (p. 503) describes it as a model that keeps “the classical dichotomy” in the long run, while in the short run “monetary policy is not neutral, owing to the assumption of price rigidity” (p. 505).

	Original Keynesian (IS–LM + Phillips curve, 1950s–60s)	New Classical (Lucas, Sargent–Wallace, Barro, 1970s)	New Keynesian (1980s on; Benigno 2015)
(i) Prices, expectations, money	Wages/prices rigid by assumption; expectations adaptive or absent; a stable, exploitable Phillips curve; money has lasting real effects.	Prices and wages flexible, markets clear; rational expectations; only <i>unanticipated</i> money moves output (misperceptions); systematic policy is ineffective.	Rational expectations <i>plus</i> pre-set prices: a fraction α keeps p^e . Policy that reacts after p^e is set moves output even if fully anticipated; neutral in the long run.
(ii) Market structure	Unmodelled or competitive; price level set by a rule of thumb or a rigid nominal wage.	Perfect competition: price-takers, no one sets a price.	Monopolistic competition (Dixit–Stiglitz): firms set $P(j) = (1 + \mu)W/A$, so $P >$ marginal cost and firms serve extra demand at a fixed price.
(iii) AD and the instrument	IS from a consumption function, LM from money demand; the central bank sets M ; AD slopes down through real balances.	Demand is the quantity equation $MV = PY$; money supply is the instrument and the source of nominal shocks.	AD is the Euler equation; no LM curve; the central bank sets i ; AD slopes down because a higher p with $\bar{p}$ fixed raises the real rate.
(iv) Micro-foundations and welfare	Behavioural equations postulated and estimated; policy judged by an ad hoc loss.	Full micro-foundations; equilibrium is efficient, so there is nothing for stabilisation policy to fix.	Micro-founded <i>and</i> distorted: the welfare loss is derived from utility, $\frac{1}{2}(y - y_e)^2 + \frac{\theta}{2\kappa}(p - p^e)^2$ (eq. 33).

Each row is expanded below, in the order the question asks: *what each school does, the evidence that pushed the NK formulation, and what it implies for the structure and predictions of the model.*

(i) Price adjustment, expectations, and the short-run effects of monetary policy

Original Keynesian.

Nominal wages (in Keynes) or prices are rigid, and the rigidity is an *assumption*. It is not derived from anyone’s choice. Expectations are either absent or adaptive, so the Phillips curve of the 1960s is read as a stable menu: more inflation buys permanently less unemployment. Monetary policy has real effects that last as long as the rigidity does.

New Classical.

Prices and wages clear markets continuously, and expectations are rational. Money moves output only through *surprises*: in Lucas’s imperfect-information model, a producer who cannot tell a general price rise from a rise in the relative price of their own good supplies more when the price level is unexpectedly high. That gives the “New-Classical Phillips curve”, $p - p^e = \kappa(y - y_n)$, of Phelps (1967) and Lucas (1973). Its policy implication is the Sargent–Wallace *policy-ineffectiveness proposition*: any systematic rule is built into expectations, so only random policy moves output, and random policy is useless.

New Keynesian.

Rational expectations stay. Price rigidity comes back, but now as a *choice*. In Benigno §4 (p. 507) a fraction α of firms must keep the price P^e they fixed *before* the shock is observed. Two justifications are given on that page. Under monopolistic competition, a firm at its optimum loses only a *second-order* amount from a small price error, so a tiny menu cost (Mankiw, 1985) is enough to rationalise not adjusting. And the same algebra describes *sticky information*. Remarkably, the resulting AS curve has exactly the New-Classical form:

$$p - p^e = \kappa (y - y_n), \quad \kappa = \frac{(1 - \alpha)(\sigma^{-1} + \eta)}{\alpha}.$$

Benigno says so openly (p. 508): the equation is the one Phelps and Lucas derived, “derived on different principles”. What changes is the *reading*:

- p^e is not a forecast that might be wrong; it is a *price already posted*. A central bank that responds to a shock *after* p^e is set moves output even if everybody knows its rule. That is the difference with policy ineffectiveness: **systematic stabilisation policy works, because it acts faster than prices can.**
- The slope is no longer a free parameter. κ is built from the share of sticky firms α and the preference parameters (σ, η) . More rigidity means a *flatter* AS curve, and in the limit $\alpha \rightarrow 0$ the AS curve is vertical and the model is classical again.
- In the long run every firm resets, the Phillips curve is vertical, and money is neutral: $\bar{p}$ appears nowhere in $\bar{y}_n$ (Benigno §4.2, p. 508).

Empirical motivation.

Three facts pushed the profession to this combination.

1. *The 1970s*. Inflation and unemployment rose together, so the stable Keynesian Phillips curve broke down. That vindicated the New-Classical insistence that expectations adjust, and every NK model keeps rational expectations for this reason.
2. *Monetary policy has large, persistent real effects* even when it is announced and data on money and prices are published within weeks. Pure misperception cannot last that long. The costly Volcker disinflation of 1979–82 is the standard example. This argues for a friction that does not depend on anyone being fooled.
3. *Micro price data* show that individual prices are changed infrequently. Benigno calibrates an average price duration of about three quarters for the United States, i.e. $\alpha \simeq 0.66$ (p. 515). That fact is what disciplines κ .

Implications for the structure and the predictions.

The model has a two-period structure with a *non-vertical* short-run AS curve and a *vertical* long-run one. Its central prediction is short-run non-neutrality together with long-run neutrality. The size of the real effect of policy is governed by $\sigma\kappa$: from the closed-form solution, $dy = -\sigma di / (1 + \sigma\kappa)$. With rigid prices, $\kappa \rightarrow 0$ and the rate cut goes entirely into output; with flexible prices, $\kappa \rightarrow \infty$ and it goes entirely into prices.

(ii) Market structure and firms’ price-setting behaviour

Original Keynesian.

Market structure is left unmodelled. Where a price equation exists it is a mark-up rule of thumb or follows from a rigid nominal wage with competitive firms. In the latter case the real wage must be *counter-cyclical*, since firms move along a falling marginal-product curve.

New Classical.

Perfect competition. Everybody takes prices as given, so the model contains no agent whose decision *is* the price. A theory of sticky prices has nobody to attach the stickiness to.

New Keynesian.

Monopolistic competition in the Dixit–Stiglitz form (Benigno eq. 9; Blanchard and Kiyotaki, 1987). Firm j faces demand $Y(j) = (P(j)/P)^{-\theta}(C+G)$, and profit maximisation gives a constant mark-up over marginal cost (eq. 11):

$$P(j) = \frac{\theta}{\theta - 1} \cdot \frac{1 + \tau_w}{1 - \tau_y} \cdot \frac{W}{A} = (1 + \tilde{\mu}) \frac{W}{A}.$$

Monopolistic competition does three separate jobs, and each is structural:

1. **Someone sets prices**, so a pre-set price P^e is a well-defined decision.
2. **Price exceeds marginal cost**, so a firm stuck at P^e is *willing* to serve extra demand. Output in the short run is therefore *demand-determined*, and that is why shifting AD moves y .
3. **The mark-up is a wedge**. It makes the natural level inefficiently low, $y_n < y_e$, and a change in the mark-up is an *inefficient* shock. This is the entire content of Questions 3 and 4.

It also explains why small frictions matter. The envelope theorem makes a firm's private loss from not adjusting second order, while the aggregate effect on output is first order, because output starts below its efficient level.

Empirical motivation.

Firms visibly set and post prices and earn mark-ups. The real wage is not strongly counter-cyclical, so a model in which all the rigidity sits in nominal wages with competitive firms predicts the wrong co-movement. Putting rigidity in *prices* set by firms with market power fixes that.

Implications.

The AS curve is a *pricing* relation: only the $1 - \alpha$ firms that can adjust move their prices, in proportion to real marginal cost, which rises with $y - y_n$ (eq. 16). Its anchor (p^e, y_n) moves with anything that changes the mark-up: monopoly power, the taxes $\tau_w, \tau_l, \tau_y, \tau_c$ in eq. (13), or an oil-price rise (p. 513). The model can therefore talk about *cost-push shocks*, which neither predecessor could.

(iii) The construction of aggregate demand and the monetary-policy instrument**Original Keynesian.**

IS comes from a consumption function in *current* income and investment in the interest rate; LM comes from money demand; the central bank sets the money supply. AD slopes down through real balances: a higher P lowers M/P , raises i and cuts investment. Consumption does not depend on the future.

New Classical.

Aggregate demand is essentially the quantity equation, with the money stock as the policy variable and the source of nominal disturbances.

New Keynesian (Benigno §3).

AD is the household's **Euler equation** combined with the resource constraint:

$$y = \bar{y}_n + (g - \bar{g}) - \sigma [i - (\bar{p} - p) - (\bar{\tau}_c - \tau_c) - \rho].$$

- **No LM curve, no money stock.** The central bank sets the short-term nominal rate i directly, and money is supplied elastically at that rate. Benigno (p. 504): the analysis “is consistent with the modern central banking practice of targeting short-term nominal interest rates, not money supply aggregates”.
- **The slope is intertemporal.** With $\bar{p}$ given, a higher p today lowers expected inflation $\bar{p} - p$, raises the real rate, and households postpone consumption. Money demand plays no role in it.
- **The future is in today's demand.** $\bar{y}_n$ and $\bar{p}$ enter directly: news about long-run income, or a promise about the long-run price level, shifts demand now. Benigno calls intertemporal decisions “another novelty with respect to Keynesian models” (p. 505).

Empirical motivation.

Central banks set interest rates; they do not target monetary aggregates, and money demand proved too unstable to make money targets reliable. The Euler-equation demand also builds the Lucas critique in, since its parameters are preferences and not reduced-form propensities.

Implications.

Transmission runs through the *real* rate. Policy works through *expectations* ($\bar{p}$ enters AD exactly where $-i$ does; see Benigno §9). The model can have a *zero lower bound*, because the instrument is a rate. And it predicts *smaller* fiscal multipliers in normal times than the Keynesian cross: a rise in g has a multiplier below one on output (Benigno §8, p. 514).

(iv) Micro-foundations and welfare analysis**Original Keynesian.**

Behavioural equations are postulated and estimated. Policy is judged by a loss function that the modeller writes down. Its targets and weights express the analyst's preferences, not the households'.

New Classical.

Full micro-foundations, but in a frictionless competitive economy. By the First Welfare Theorem the equilibrium is efficient, so fluctuations are optimal responses and there is nothing for stabilisation policy to fix.

New Keynesian.

Micro-foundations *with* distortions: market power, taxes and sticky prices. Because the model has a household utility function and identifiable distortions, the objective of policy is *derived*. A second-order approximation to utility around an efficient steady state gives Benigno's eq. (33):

$$L = \frac{1}{2} (y - y_e)^2 + \frac{\theta}{2\kappa} (p - p^e)^2.$$

This loss says three things no ad hoc loss can justify. The output target is the *efficient* level y_e , not the natural level. Only *unanticipated* price movements are costly, because with a share α of prices frozen they create relative-price dispersion across identical goods. And the weight on

prices, θ/κ , rises with rigidity and with substitutability. Welfare analysis then answers questions a Keynesian model cannot even pose. Which shocks create a trade-off? Only mark-up shocks, as Question 3 shows. Does a positive fiscal multiplier mean the policy is good? Not necessarily: Benigno §8 shows that a cut in the consumption tax raises output but has a *negative* multiplier on the output gap.

Empirical motivation.

Large macro-econometric models broke down out of sample in the 1970s. The Lucas critique says why: their coefficients shift when policy changes. Deep parameters $(\beta, \tilde{\sigma}, \eta, \theta, \alpha)$ are the defence.

Implications.

κ , σ , y_n and y_e are functions of deep parameters. Policy rankings follow from utility. The model delivers the “divine coincidence” for efficient shocks and a genuine trade-off for inefficient ones, and it rationalises *flexible inflation targeting* (Benigno §11). The model’s limits, stated by Benigno himself in §12: it has two periods only, no inflation dynamics (the AS is in the price level), no financial intermediation and no open economy.

Traps in Question 1

- **“New Keynesian = Keynesian with micro-foundations.”** Half right. The NK model keeps the New Classicals’ rational expectations and long-run neutrality; what it takes from Keynes is only that nominal prices do not clear markets in the short run.
- **Calling Benigno’s AS curve a New-Keynesian Phillips curve.** It is the New-Classical form in the price level (p. 508, fn. 7). The forward-looking Calvo curve is outside this course.
- **Justifying AD’s slope with money demand.** There is no LM curve. The slope is the real-rate channel of the Euler equation.
- **Treating the welfare loss as a modelling choice.** In the NK model it is *derived*; that is the point of dimension (iv).

Where the marks are

3 points, roughly 0.75 per dimension. In each dimension the marks are for:

- stating the three positions correctly;
- naming one empirical fact that motivated the NK version;
- giving one structural implication (a slope, an instrument, a target).

A comparison that lists features without saying *why* the NK model changed them earns about half.

Question 2 Two shocks, two diagrams

Method: the plotting rule, then the closed form

1. Does the change move y_n ? If so, redraw AS through the *new anchor* (p^e, y'_n) with the same slope κ . AS always passes through (p^e, y_n) .
2. Does it move AD? Only if it changes i , g , τ_c , $\bar{p}$ or the future $(\bar{y}_n)$.
3. Read the new equilibrium, and **report both gaps**: $y - y_n$ drives prices, $y - y_e$ drives welfare.

Solving AS and AD together (Benigno §5; derived in Map/benigno/04) gives the only formulas needed, with $g = \bar{g}$, no tax changes and $\bar{p} = p^e$:

$$y - y_n = \frac{\sigma}{1 + \sigma\kappa} [r_n - i], \quad p - p^e = \kappa (y - y_n), \quad r_n \equiv \rho + \sigma^{-1} (\bar{y}_n - y_n). \quad (1)$$

Start at E : $i = r_n = \rho$, $y = y_n = y_e = 0$, $p = p^e = \bar{p}$.

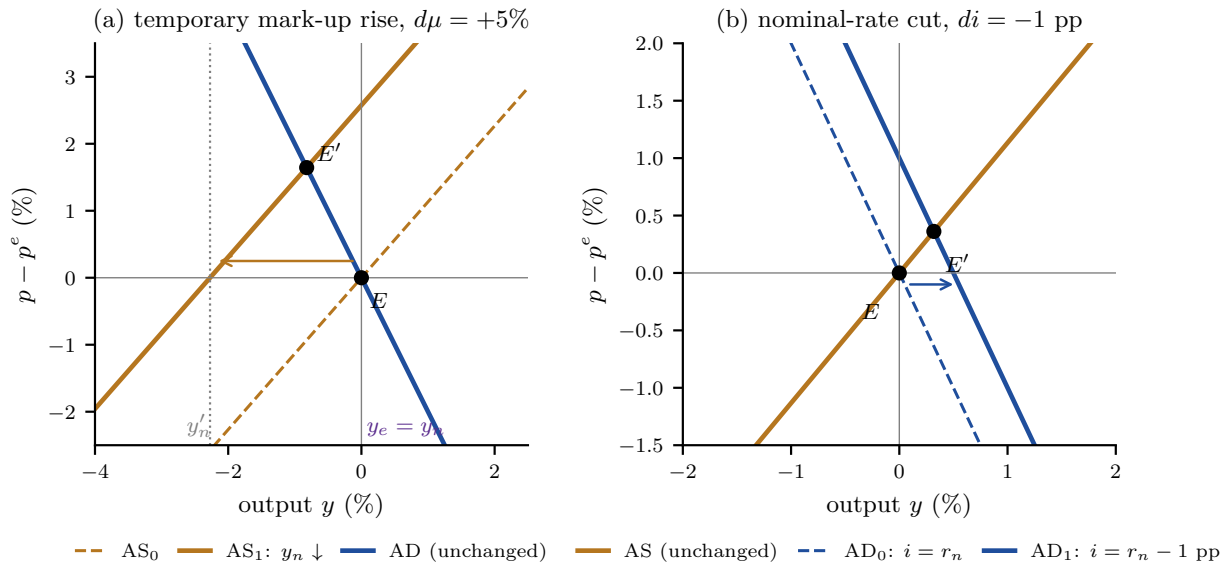


Figure 1: Question 2 in Benigno's calibration. (a) A temporary 5% rise in the desired mark-up: AS shifts up and to the left through (p^e, y'_n) ; AD does not move. (b) A one-point cut in i : AD shifts up by one point (right by $\sigma = 0.5$ points); AS does not move. Interactive version: [Two Shocks](#).

Item (a) — a temporary increase in firms' desired mark-ups

Which curve shifts, and which way.

A higher desired mark-up means a higher μ in eq. (13), whether it comes from more monopoly power ($\theta \downarrow$), higher τ_w, τ_y, τ_l , or a costlier inelastically-demanded input such as oil (p. 513). By eq. (15):

$$dy_n = -\frac{d\mu}{\sigma^{-1} + \eta} < 0, \quad dy_e = 0.$$

AS shifts up and to the left, through (p^e, y'_n) with $y'_n < y_n$. The vertical shift at a given output is

$$dp|_y = -\kappa dy_n = \frac{1 - \alpha}{\alpha} d\mu.$$

The $(1 - \alpha)/\alpha$ factor is worth reading. The adjusting firms raise their price *relative to the index* by $d\mu$, and the index rises with them.

AD does not shift. The shock is *temporary*, so $\bar{\mu}$ and hence $\bar{y}_n$ are unchanged, and μ does not appear in eq. (21). (A *permanent* rise would also move AD down through $\bar{y}_n$; that is a different question.)

Effects on output and prices.

With i unchanged at ρ , and since r_n rises by $-dy_n/\sigma$, eq. (1) gives

$$dy = \frac{\sigma\kappa}{1 + \sigma\kappa} dy_n < 0, \quad dp = -\frac{\kappa}{1 + \sigma\kappa} dy_n > 0.$$

Output falls and the price level rises: stagflation. With $d\mu = +5\%$: $dy_n = -2.27\%$, $dy = -0.82\%$, $dp = +1.64\%$.

The two gaps have opposite signs.

$$y - y_n = -\frac{dy_n}{1 + \sigma\kappa} = +1.45\% > 0, \quad y - y_e = dy = -0.82\% < 0.$$

Output is *above* the new natural level, which is why prices are rising: by AS, prices move only with $y - y_n$. At the same time it is *below* the efficient level, which is why welfare has fallen.

Economic reading. The adjusting firms want a larger margin over marginal cost, so at any output they charge more; that is the upward shift of AS. The price level rises. With $\bar{p}$ anchored, a higher p means lower expected inflation $\bar{p} - p$ and so a higher real rate $i - (\bar{p} - p)$. Households save more and bring consumption down. Demand, and with it output, falls along the unchanged AD curve. Output does not fall all the way to y'_n . The firms with frozen prices keep selling at p^e , and as output falls, marginal cost falls with it, which restrains the adjusting firms. The equilibrium E' splits the shock between prices and quantities in the ratio set by $\sigma\kappa$.

Economic reading. Policy relevance, previewing Question 4. No interest rate restores both targets. Holding $p = p^e$ requires *raising* i to r_n (to 6.55% here) and pushes output down to y'_n . Holding $y = y_e$ requires *cutting* i and accepts a price rise of 2.58%. Because μ moved y_n without moving y_e , this is the model's genuine trade-off.

Item (b) — a reduction in the nominal interest rate by the central bank

Which curve shifts, and which way.

i appears only in AD. Solving eq. (21) for p ,

$$p = \bar{p} + \rho - i + \frac{\bar{y}_n - y}{\sigma},$$

so a cut $di < 0$ shifts **AD up by $|di|$ at every output, equivalently to the right by $\sigma|di|$** . **AS does not move:** y_n depends on a , g and μ , none of which the central bank touches, and p^e is predetermined.

Effects on output and prices.

From eq. (1), with r_n unchanged:

$$dy = -\frac{\sigma}{1 + \sigma\kappa} di > 0, \quad dp = \kappa dy > 0.$$

Output and the price level both rise, moving up along AS. With $di = -1$ pp: $dy = +0.32\%$, $dp = +0.36\%$. Since neither y_n nor y_e moved, both gaps equal $+0.32\%$. Starting from $i = r_n$, the cut *overheats* the economy: output rises above both its natural and its efficient level.

Economic reading. At a given p , and with $\bar{p}$ anchored, a lower nominal rate is a lower *real* rate. Households substitute toward consumption today (the Euler equation), so demand rises at every price level. The firms with frozen prices (α) serve that demand at p^e , so output rises. Hiring more labour raises the real wage households require, and with it real marginal cost, so the $1 - \alpha$ adjusting firms raise their prices. The price level rises, which partly offsets the fall in the real rate. That feedback is why the output response $\sigma/(1 + \sigma\kappa)$ is smaller than the horizontal shift of AD, σ .

Economic reading. What does not happen. There is no liquidity effect through a money market and no LM curve: the central bank sets i directly. And the effect is purely short run. In the second period every firm resets, $\bar{y} = \bar{y}_n$, and only $\bar{p}$ could be affected: money is neutral

in the long run (Benigno §4.2). A cut is the right policy only when r_n has fallen, for example after a temporary productivity gain (§6.1) or a fall in $\bar{y}_n$. In that case it restores E instead of overheating.

Traps in Question 2

- **Shifting AD in (a).** A *temporary* mark-up shock leaves $\bar{y}_n$ unchanged; only AS moves. The fall in demand is a movement *along* AD caused by the higher price level.
- **Reporting one “output gap”.** After (a), $y - y_n > 0$ while $y - y_e < 0$. The first explains inflation, the second welfare.
- **More rigidity makes AS steeper.** It is the opposite: $\alpha \uparrow \Rightarrow \kappa \downarrow$, flatter AS, and a rate cut moves output more and prices less.
- **Using the IS–LM mechanism in (b)** (money supply up, liquidity effect). There is no LM curve; the channel is the real rate in the Euler equation.

Where the marks are

1.5 points per item. For each item the marks are for:

- the correct curve and direction (and, in (a), saying that AD does *not* move);
- the signs of dy and dp ;
- the transmission mechanism in words (mark-up $\rightarrow$ prices $\rightarrow$ real rate $\rightarrow$ demand; or rate $\rightarrow$ real rate $\rightarrow$ demand $\rightarrow$ marginal cost $\rightarrow$ prices).

Distinguishing the two gaps in (a) is what separates full marks from most answers.

Question 3 Natural versus efficient output

Definitions.

- **Natural output y_n :** the output of the *decentralised* economy when *every* firm can reset its price. It is the flexible-price equilibrium of an economy that still has monopolistic competition and distortionary taxes. It is the anchor of AS, and $y - y_n$ drives *prices*.
- **Efficient output y_e :** the output a *benevolent planner* would choose, maximising $u(C) - v(L)$ subject to $Y = C + G$ and $Y = AL$. It is the target in the welfare loss, and $y - y_e$ drives *welfare*.

Neither concept involves price stickiness: α appears in neither. Stickiness explains why y differs from y_n . The mark-up explains why y_n differs from y_e .

Step 1. The planner’s condition.

Substituting both constraints, the planner solves $\max_Y u(Y - G) - v(Y/A)$. The first-order condition is

$$\frac{v_l(Y_e/A)}{u_c(Y_e - G)} = A : \quad \text{MRS} = \text{MRT}.$$

The marginal rate of substitution between leisure and consumption equals the marginal rate of transformation, which is labour productivity.

Step 2. The market’s condition.

With flexible prices every firm sets $P = (1 + \mu)W/A$, so the real wage is $W/P = A/(1 + \mu)$. The

household equates its MRS to that real wage (eq. 14):

$$\frac{v_l(Y_n/A)}{u_c(Y_n - G)} = \frac{A}{1 + \mu}.$$

The two conditions are identical except for the wedge $1 + \mu$.

Step 3. Log-linearise and subtract.

With $v_l = L^\eta$, $u_c = C^{-\tilde{\sigma}^{-1}}$ and $\tilde{\sigma}^{-1}c = \sigma^{-1}(y - g)$, the log of the MRS is $\eta(y - a) + \sigma^{-1}(y - g)$. The planner sets it equal to a , the market to $a - \mu$. This gives eqs. (19) and (15), and

$$y_n - y_e = -\frac{\mu}{\sigma^{-1} + \eta}$$

(Benigno §4.4, p. 509).

Why they differ under monopolistic competition.

Market power drives a wedge between the marginal product of labour and the real wage. A firm facing downward-sloping demand restricts output to keep its price above marginal cost. Equivalently, it pays labour only $A/(1 + \mu)$ of what labour produces. The household works until its MRS equals the wage it is paid, not the product it creates. So it works, and the economy produces, *too little*. At y_n the value to society of one more unit of output (MRT = A) exceeds its cost to the household (MRS = $A/(1 + \mu)$), and an extra unit would raise welfare. The shaded triangle in Figure 2(i) is that lost surplus. Its area grows with the square of the distance, $\frac{1}{2}(\sigma^{-1} + \eta)(y_n - y_e)^2$, which is why the welfare loss is quadratic in $y - y_e$. Taxes act the same way, since every tax in eq. (13) is part of μ . In a perfectly competitive, tax-free economy ($\theta \rightarrow \infty$, all $\tau = 0$), $\mu = 0$ and the two concepts coincide.

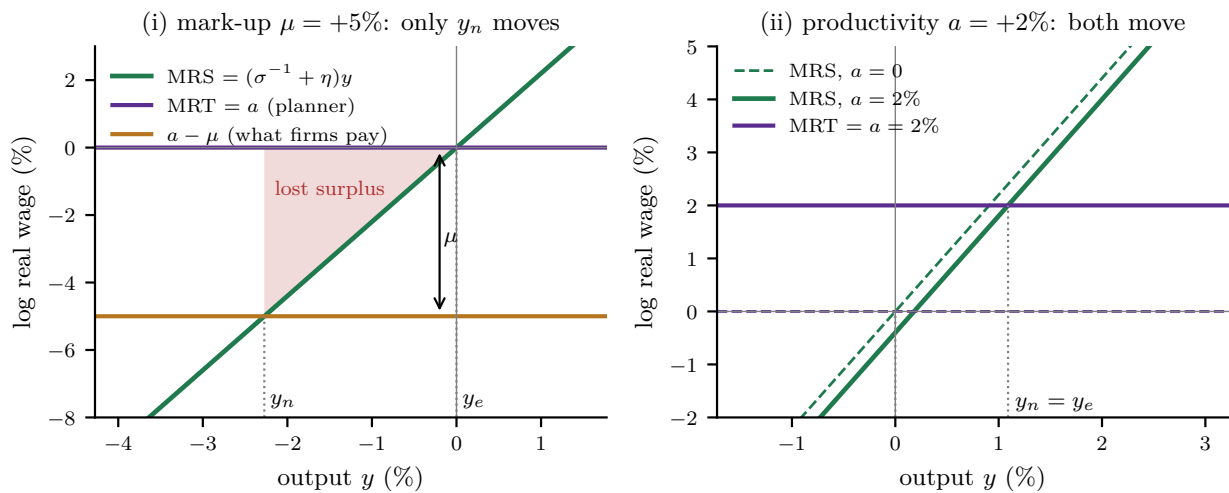


Figure 2: Question 3 in logs, Benigno's calibration ($\sigma^{-1} + \eta = 2.2$). (i) A 5% mark-up lowers the wage firms pay, not the product of labour. The market stops at $y_n = -2.27\%$ and the planner at $y_e = 0$. (ii) A 2% productivity gain raises both the MRT and the MRS curve's intercept, so y_n and y_e rise together to 1.09%. Interactive version: [The Wedge](#).

Why a temporary rise in the desired mark-up changes y_n but not y_e .

Look at what enters each problem:

$$y_n = y_n(a, g, \mu), \quad y_e = y_e(a, g).$$

The planner's problem contains **preferences, technology and resources** (u, v, A, G) and nothing else. It contains no prices, no market power and no taxes. A mark-up is not a change in what the economy *can* produce or what households *want*. It is a change in how the *market* allocates: a larger wedge between price and marginal cost. So

$$dy_n = -\frac{d\mu}{\sigma^{-1} + \eta} < 0, \quad dy_e = 0.$$

The mark-up is an inefficient shock: it moves the economy's flexible-price outcome without moving its welfare target. "Temporary" adds only that $\bar{\mu}$, and hence $\bar{y}_n$ and AD, stay put; y_e would not move at any horizon.

Compare a productivity shock, Figure 2(ii). It moves MRT and MRS by the same amount in both problems, so $dy_n = dy_e$. That is an *efficient* shock. The distinction is the whole of Benigno's policy analysis:

a policy trade-off exists $\iff$ the shock moves $y_n - y_e$ $\iff$ the shock moves μ .

Economic reading. Why the distinction matters for policy. Monetary policy can close $y - y_n$, and by doing so stabilise prices. When $y_n = y_e$ that also closes the welfare gap: the "divine coincidence" of Benigno §6. After a mark-up shock the two targets separate. Any interest rate trades one against the other, and no interest rate can move y_n back to y_e : that would require removing the wedge itself, through tax cuts or competition policy.

Traps in Question 3

- **"Natural" = "optimal".** y_n is the flexible-price outcome of a *distorted* economy. It is where prices are stable, not where welfare is highest.
- **Attributing $y_n \neq y_e$ to sticky prices.** Neither concept depends on α . The gap comes from market power and taxes alone.
- **Saying the mark-up lowers y_e because "firms produce less".** Firms do produce less, and that is y_n falling. The planner's optimum has not moved, which is exactly why the fall is a welfare loss.

Where the marks are

2 points. The marks are for:

- the two definitions (flexible-price decentralised outcome versus planner's optimum);
- the wedge argument ($MRS = A/(1 + \mu)$ against $MRS = A$), ideally with $y_n - y_e = -\mu/(\sigma^{-1} + \eta)$;
- the reason μ is absent from the planner's problem.

A contrast with a productivity shock is the cleanest way to show the last point.

Question 4 The loss function and the AS constraint

Map to Benigno's notation

The question's loss is a general version of Benigno's eq. (33). Setting

$$\phi_y = \frac{1}{2}, \quad \phi_p = \frac{\theta}{2\kappa}, \quad y^* = y_e$$

recovers (33) exactly; setting $\phi_p = \phi/(2\kappa)$ with a free ϕ recovers the mandate version (35). Everything below is solved for general (ϕ_y, ϕ_p, y^*) and then specialised.

The shape of the loss function, and what it says about the central bank

$$\mathcal{L} = \phi_y (y - y^*)^2 + \phi_p (p - p^e)^2, \quad \phi_y, \phi_p > 0.$$

1. **A bliss point.** The loss is zero only at $(y, p) = (y^*, p^e)$. The bank has *two* targets: output at y^* and no price surprise.
2. **Quadratic, so convex and symmetric.** Deviations in either direction are equally bad. A boom above y^* is penalised as much as a recession below it, and a price rise as much as a fall. The bank is not trying to push output up; it *stabilises* around a target. Convexity means the marginal loss rises with the size of the deviation: small misses are cheap and large ones very costly. So when the two targets conflict, the bank *spreads* the miss across both instead of hitting one exactly and ignoring the other.
3. **Iso-loss curves are ellipses** centred at the bliss point (Figure 3(i)). Their axis ratio is $\sqrt{\phi_y/\phi_p}$. A large ϕ_p/ϕ_y gives flat, wide ellipses: the bank tolerates output deviations far more than price surprises.
4. **The relative weight ϕ_p/ϕ_y is the bank's conservatism.** A large value is a hawk, a small one a dove. In the welfare-based version, $\phi_p/\phi_y = \theta/\kappa$, so the household *itself* wants more weight on prices when goods are more substitutable ($\theta \uparrow$) or prices more rigid ($\alpha \uparrow \Rightarrow \kappa \downarrow$). The reason is that each price surprise then creates more costly relative-price dispersion (Benigno p. 522).
5. **The price term is $p - p^e$, not p .** Only *unanticipated* movements are costly, because only they create dispersion between the firms that adjusted and those stuck at p^e . An anticipated price level is costless in this model.
6. **The output term uses y^* .** In the welfare-based loss $y^* = y_e$, not y_n . The bank cares about the allocation households experience, not about firms' marginal cost.

Why the problem is subject to the AS condition

The bank has **one instrument**, i , and it appears only in AD. Moving i slides AD along the AS curve, so the bank can reach *any* point on AS, and *only* points on AS. The AS curve is private-sector behaviour that policy cannot override within the period. The $1 - \alpha$ adjusting firms price off marginal cost, the α others are stuck at p^e , and p^e was set before the shock.

So AS is the central bank's *feasible set*. It is its opportunity frontier, exactly as a budget line is a consumer's. AD is not a constraint: it contains the free variable i , so for any point on AS there is an i that puts AD through it. Choosing (y, p) subject to AS and then backing out i is the same as choosing i directly.

Solving it.

The Lagrangian is

$$\Lambda = \phi_y (y - y^*)^2 + \phi_p (p - p^e)^2 + \lambda [p - p^e - \kappa(y - y_n)].$$

Its first-order conditions are

$$2\phi_y (y - y^*) - \lambda\kappa = 0, \quad 2\phi_p (p - p^e) + \lambda = 0.$$

Eliminating λ gives the **targeting rule**:

$$\boxed{\phi_y (y - y^*) + \kappa \phi_p (p - p^e) = 0} \tag{2}$$

Geometrically, (2) is the tangency of an iso-loss ellipse with AS. The marginal rate of substitution between the targets, $-\phi_y(y - y^*)/[\phi_p(p - p^e)]$, equals the marginal rate of transformation κ along AS. With Benigno's weights it becomes $(y - y_e) + \theta(p - p^e) = 0$, his eq. (34), the IT line.

Substituting AS into (2) gives

$$y^o = \frac{\phi_y y^* + \phi_p \kappa^2 y_n}{\phi_y + \phi_p \kappa^2}, \quad p^o - p^e = \frac{\kappa \phi_y}{\phi_y + \phi_p \kappa^2} (y^* - y_n), \quad \mathcal{L}^o = \frac{\phi_y \phi_p \kappa^2}{\phi_y + \phi_p \kappa^2} (y^* - y_n)^2. \quad (3)$$

Optimal output is a *weighted average* of the target y^* and the natural level y_n . Everything hinges on one number, $y^* - y_n$: the distance between where the bank wants output and where the AS curve crosses $p = p^e$. The bank implements y^o with the rate that puts AD through it. From eq. (1), $i^o = r_n - (1 + \sigma\kappa)(y^o - y_n)/\sigma$.

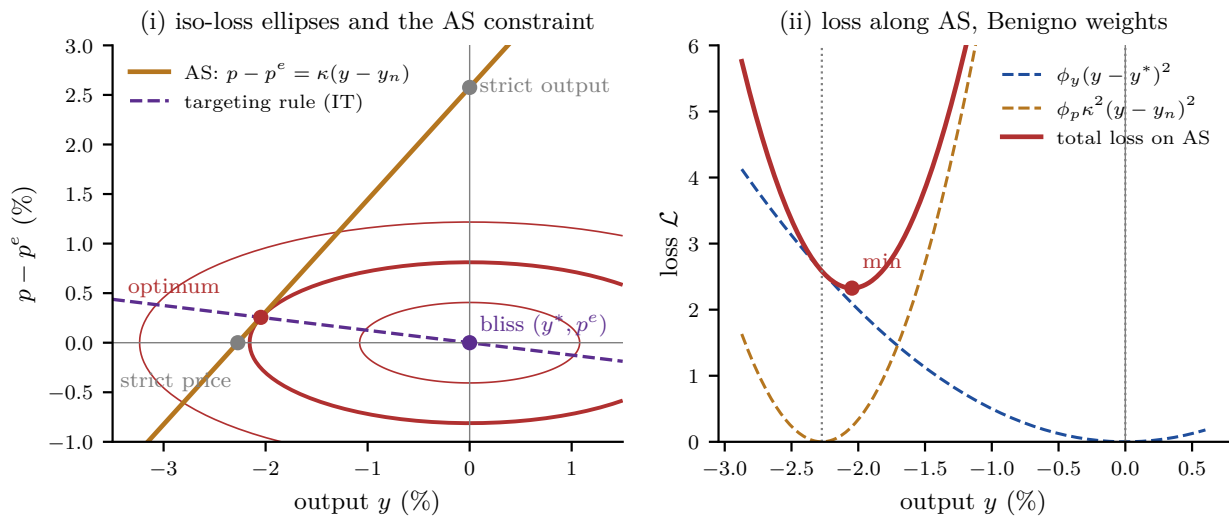


Figure 3: Question 4 after a 5% mark-up shock, with Benigno's weights ($\phi_y = \frac{1}{2}$, $\phi_p = \theta/2\kappa = 3.53$, $y^* = y_e = 0$). (i) The optimum is where an iso-loss ellipse is tangent to AS. It lies on the targeting rule, between strict price stability and strict output stabilisation. (ii) The same problem along AS: the output part falls and the price part rises as y moves from y_n toward y^* , and the total is minimised in between. Interactive version: [The Loss Bowl](#).

What the AS constraint means for economic policy

1. When $y^* = y_n$, the constraint does not bind: the divine coincidence.

If the shock moves y_n and $y^* = y_e$ together, as productivity and public spending do, then AS passes through the bliss point. Eq. (3) gives $y^o = y^*$, $p^o = p^e$ and $\mathcal{L}^o = 0$. **Stabilising prices and stabilising the welfare-relevant gap are the same policy**, and the bank sets $i = r_n$.

2. When $y^* \neq y_n$, the constraint binds: a genuine trade-off.

After a mark-up shock y_n falls and y_e does not, so AS misses the bliss point and $\mathcal{L}^o > 0$. With Benigno's weights and $y^* = y_e$, the optimum is

$$p^o - p^e = \frac{1}{1 + \theta\kappa} \cdot \frac{d\mu}{\sigma^{-1} + \eta}, \quad y^o - y_e = -\frac{\theta\kappa}{1 + \theta\kappa} \cdot \frac{d\mu}{\sigma^{-1} + \eta}.$$

The bank lets a fraction $1/(1 + \theta\kappa)$ of the shock into prices and absorbs the rest as an output loss. With $\theta\kappa = 9.07$ that fraction is 9.9%. For $d\mu = +5\%$: $y^o = -2.05\%$ and $p^o - p^e = +0.26\%$. The implementing rate is $i^o = 5.84\%$: a rise from 2%, but less than the 6.55% ($= r_n$) that strict price stability would need. The two limits are the corners of Figure 3(i):

$\phi_p/\phi_y \rightarrow \infty$ is strict price stability ($y = y_n$, $p = p^e$); $\phi_p/\phi_y \rightarrow 0$ is strict output stabilisation ($y = y^*$, $p - p^e = 2.58\%$). Even the *sign* of the response relative to doing nothing depends on the mandate (p. 522). A bank less concerned with prices, whose IT line is steeper than AD, would *cut* rates after the same shock.

3. Monetary policy chooses a point on AS; it cannot move AS.

The constraint encodes what the central bank *cannot* do. It cannot change y_n , so it cannot close $y_n - y_e$. The permanent part of that gap, the level distortion from market power and taxes, is a job for *structural and fiscal* policy: a subsidy or a cut in τ_w , τ_l or τ_y that lowers μ , or competition policy that raises θ . Asking monetary policy to deliver $y^* > y_n$ *permanently* is asking it to beat a constraint it does not control.

4. The trade-off exists only because prices are sticky, and only in the short run.

The slope of the frontier is κ . As $\alpha \rightarrow 0$, $\kappa \rightarrow \infty$ and AS becomes vertical at y_n : the bank can choose p but not y , and the optimum is $p = p^e$. In the second period every firm resets, AS is vertical, and policy only sets $\bar{p}$. Moreover p^e is predetermined *within* the period, not forever. If the bank tried to hold y above y_n every period, price-setters would build that into p^e . Surprises cannot be systematic: with rationally set p^e they average zero, so $y = y_n$ on average whatever the bank wants. This is the time-inconsistency logic of Kydland–Prescott and Barro–Gordon, and it is why Benigno approximates welfare around an *efficient* steady state (fn. 21, p. 522). There $y_n = y_e$ absent shocks, so the bank has no permanent incentive to surprise.

5. A rule the bank can follow without knowing the shock.

Eq. (2) does not mention the shock. A bank that keeps $\phi_y(y - y^*) + \kappa\phi_p(p - p^e) = 0$ responds optimally to productivity, spending and mark-up shocks, whether temporary, permanent or expected. This is Benigno’s argument (pp. 521–522, after Giannoni and Woodford, 2002) for a *targeting rule* over a Taylor rule, whose optimal coefficients depend on the shock process. It is also what “flexible inflation targeting” means operationally (Svensson): let prices deviate only in proportion to the output sacrifice the AS curve would otherwise force.

Traps in Question 4

- **Using AD as the constraint.** AD contains the instrument, so it restricts nothing. AS is the constraint because it is the one relation policy cannot shift.
- **Setting $y^* = y_n$ in the welfare-based loss.** The welfare target is y_e . With $y^* = y_n$ there is never a trade-off, and the question loses its point.
- **“The bank minimises inflation.”** It minimises price *surprises* $p - p^e$, jointly with output deviations. The weights decide how the unavoidable miss is split.
- **Reading κ as a policy choice.** It is the private sector’s pricing technology. The bank takes it as given, exactly as a consumer takes prices as given.

Where the marks are

2 points. The marks are for:

- the shape: bliss point, convexity and symmetry, and the relative weight as the bank’s conservatism;
- the reason AS is the constraint (one instrument, which moves AD only; AS is the feasible set);
- the policy meaning: no trade-off when $y^* = y_n$, a trade-off after mark-up shocks, and monetary policy’s inability to move y_n .

The targeting rule (2) and the split $1/(1 + \theta\kappa)$ are the full-marks extension.